

## MATH X313 PRACTICE EXAM

### Instructions:

1. Do all questions.
2. You have 2 hours to complete this exam.
3. You are allowed 2 double-sided A4 paper handwritten notes.
4. You may use any type of calculator.
5. In the exam venue, you will be given an answer booklet. Write your answers in the answer booklet. It is not necessary to copy the question again in the answer booklet.
6. Anything written on the question paper itself will not be marked.

### Practice Exam Questions

1. In this question,  $\{Z_t\}$  is white noise  $WN(0, \sigma^2)$  and  $B$  is the backward shift operator. Determine whether the following processes are causal and/or invertible. You need to show the full working to obtain full credit for each part.

- (a)  $X_t = 2X_{t-2} + Z_t + 0.25Z_{t-2}$ .
- (b)  $X_t = 0.36X_{t-4} + Z_t - 0.25Z_{t-1}$ .
- (c)  $X_t = -0.49X_{t-2} + Z_t - 1.4Z_{t-2}$ .
- (d)  $X_t = 0.36X_{t-1} + Z_t - 0.49Z_{t-4}$ .

[4 + 4 + 4 + 4 = 16 marks]

2. We consider the time series model

$$(1 - 0.5B)(1 - B)X_t = (1 + 0.25B)Z_t,$$

where  $Z_t$  is white noise  $WN(0, 0.01)$ , in this question.

- (a) Is  $\{X_t\}$  a stationary time series? Provide your reason.
- (b) Show that we can rewrite  $Z_t$  in the form  $Z_t = \sum_{j=0}^{\infty} a_j X_{t-j}$  where  $\sum_{j=0}^{\infty} |a_j| < \infty$  and calculate  $a_0$ ,  $a_1$  and  $a_2$ .
- (c) Suppose  $X_n$  is the last term of the series and we have

$$X_n = 0.2, X_{n-1} = 0.18, X_{n-2} = 0.19, \text{ and } Z_n = 0.009, Z_{n-1} = 0.011.$$

Calculate  $\hat{X}_{n+1}$  and  $\hat{X}_{n+2}$  and their respective error variances.

[2 + 8 + 6 = 16 marks]

3. In Figure 1, we have a time plot of the lynx trapping data from 1821 to 1934. In Figure 2, we have the plot of the same data in natural log scale. In Figure 3, we

Figure 1: Plot of lynx data

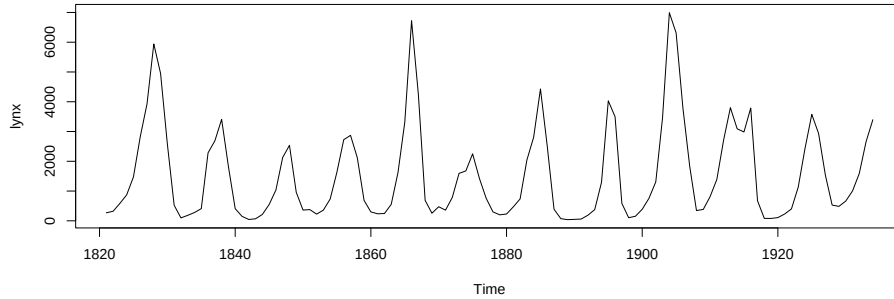
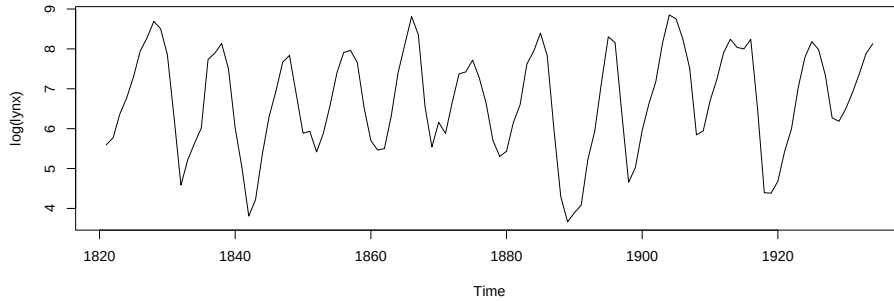


Figure 2: Plot of lynx data in natural log scale



have the periodogram of  $\log(\text{lynx})$ , where the frequency that corresponds to the higher spectral value is  $f = 0.1083333333$ . The autocorrelogram and partial autocorrelogram of  $\log(\text{lynx})$  are shown in Figures 4 and 5 respectively. The partial autocorrelogram

Table 1: List of non-seasonal  $\log(\text{lynx})$  models and their AIC values

|       |         |         |          |          |
|-------|---------|---------|----------|----------|
| ARIMA | (8,0,3) | (9,0,2) | (10,0,1) | (11,0,0) |
| AIC   | 173.56  | 171.69  | 169.38   | 166.13   |

of  $\log(\text{lynx})$  deseasonalised with period 10 is shown in Figure 6.

- Looking at the plot of the data in original scale, is it heteroscedastic or not? Provide your reasons. If it is, name some transformations that you can apply to the data in the original scale.
- If you perform the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test to  $\log(\text{lynx})$ , what result do you expect to see?

Figure 3: Periodogram of  $\log(\text{lynx})$

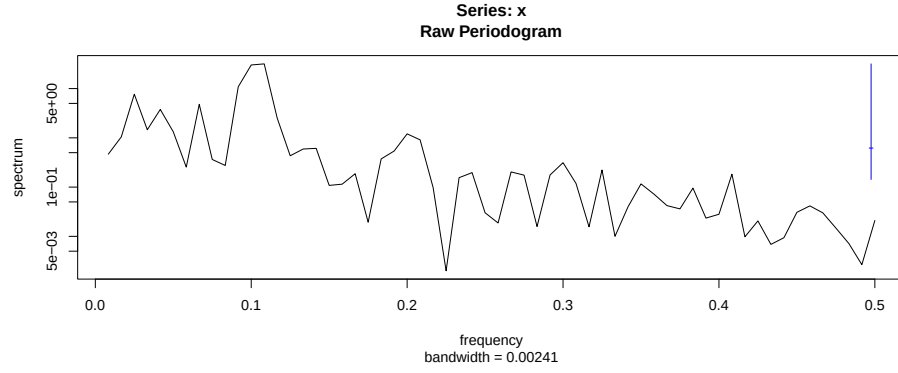


Figure 4: Autocorrelogram of  $\log(\text{lynx})$

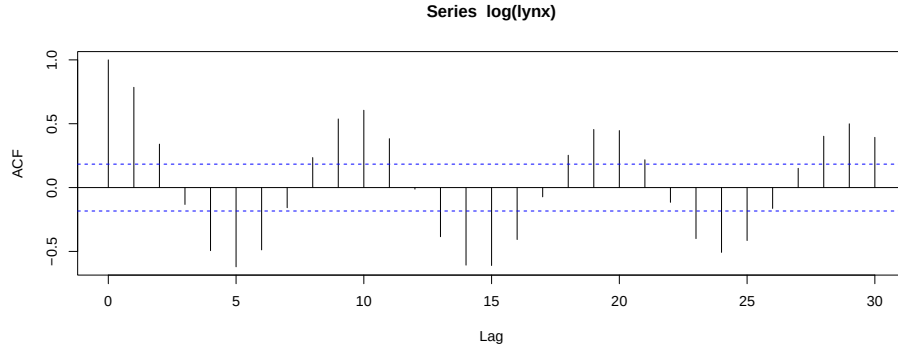


Figure 5: Partial autocorrelogram of  $\log(\text{lynx})$

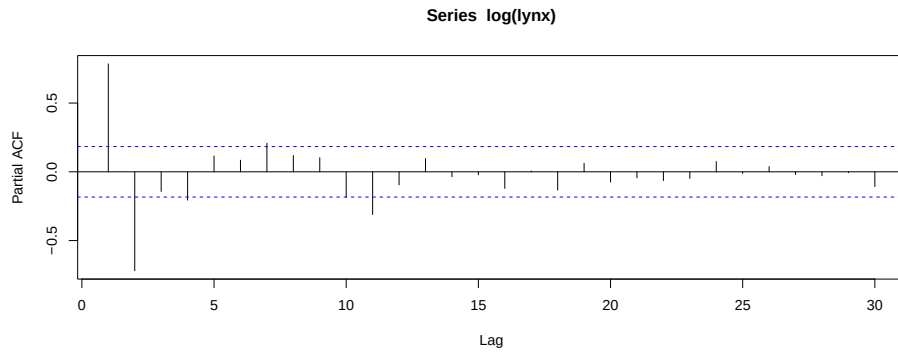


Table 2: List of seasonal  $\log(\text{lynx})$  models and their AIC values

|        |                      |                      |
|--------|----------------------|----------------------|
| SARIMA | $(2,0,1)(1,1,0)[10]$ | $(2,0,0)(1,1,0)[10]$ |
| AIC    | 172.53               | 170.93               |

Figure 6: Partial autocorrelogram of deseasonalised  $\log(\text{lynx})$  (period 10)

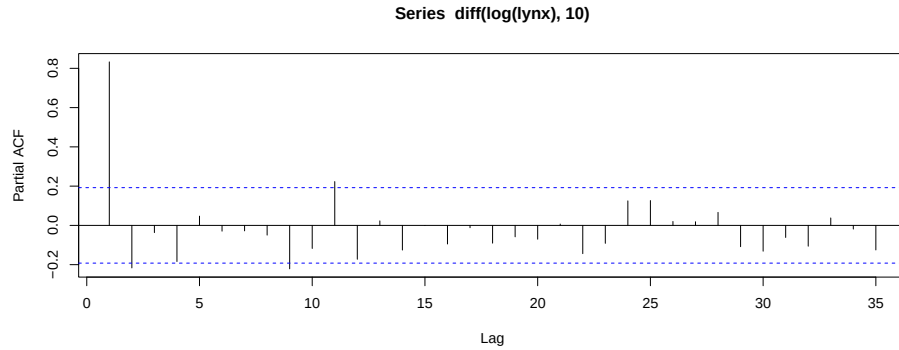
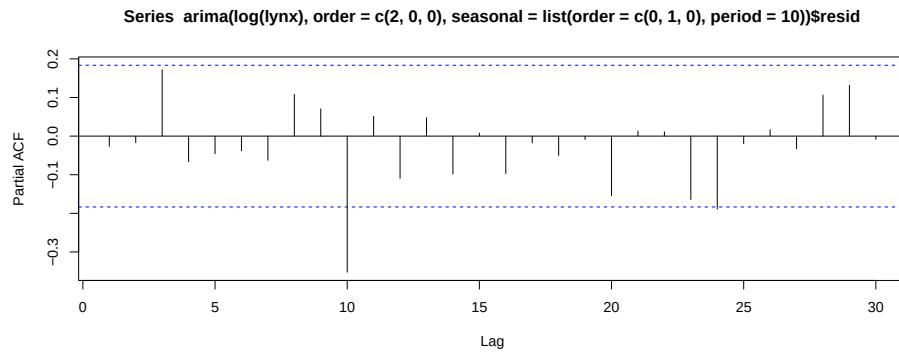


Figure 7: Partial autocorrelogram of residuals of SARIMA(2,0,0)(0,1,0)[10] model of  $\log(\text{lynx})$



- (c) Looking at the autocorrelogram and partial autocorrelogram of  $\log(\text{lynx})$ , what could be a possible viable initial model if you chose to fit a non-seasonal model? Provide your reasons.
- (d) Suppose you have obtained a viable initial non-seasonal model, how could you look for a few more other viable non-seasonal models?
- (e) The models listed in Table 1 all have white noise residuals, one of which was the initial viable non-seasonal model. Which model would you choose and why?
- (f) If you want to work with a seasonal model instead, what periods can you would choose for the seasonal part?
- (g) From Figure 6, the partial autocorrelogram of the deseasonalised  $\log(\text{lynx})$  ( $((1 - B^{10}) \ln(X_t))$ ), what could be your initial seasonal model for  $\log(\text{lynx})$ ? (Significant spikes are at lags 1,2,9,11.)
- (h) The partial autocorrelogram of the residuals of the seasonal model SARIMA(2,0,0)(0,1,0)[10] is shown in Figure 7. What further adjustments could you make to try to obtain a seasonal model with white noise residuals?

- (i) Two seasonal models with white noise residuals and their respective AIC are shown in Table 2. Considering also Table 1, which model would you choose? Provide your reasons.

- (j) The summary output for the SARIMA(2,0,0)(0,1,0)[10] is shown below.

Call:

```
arima(x = log(lynx), order = c(2, 0, 0), seasonal = list(order =
  c(1, 1, 0), period = 10))
```

Coefficients:

|      | ar1    | ar2     | sar1    |
|------|--------|---------|---------|
|      | 1.1549 | -0.3831 | -0.4621 |
| s.e. | 0.0950 | 0.0971  | 0.0939  |

sigma^2 estimated as 0.2701: log likelihood = -81.46, aic = 170.93

Write down the equation of this model and verify the calculation of its AIC.

- (k) Is a GARCH model appropriate for modelling log(lynx)?

[4 + 2 + 3 + 3 + 3 + 2 + 3 + 4 + 4 + 4 + 3 = 35 marks]

4. Consider a 2-dimensional vector time series  $\mathbf{x}_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix}$ , where

$$(\mathbf{I} - \mathbf{A}B^2)(\mathbf{I} - \mathbf{I}B)\mathbf{x}_t = \mathbf{z}_t,$$

where  $\mathbf{I}$  is the  $2 \times 2$  identity matrix,  $\mathbf{A} = \begin{pmatrix} 0.1 & 0 \\ 0.02 & 0.2 \end{pmatrix}$ ,  $B$  is the backshift operator and  $\mathbf{z}_t$  are independently and identically distributed multivariate Gaussian  $\text{WN}(\mathbf{0}, \Sigma)$ , where  $\Sigma = \begin{pmatrix} 0.1 & 0.01 \\ 0.01 & 0.1 \end{pmatrix}$  is the error variance-covariance matrix.

- (a) What can you say about the behaviour of the components of the series  $\{\mathbf{x}_t\}$ ?  
 (b) The last 3 observations of the series are

$$\mathbf{x}_{n-2} = \begin{pmatrix} 1.25 \\ 0.9 \end{pmatrix}, \mathbf{x}_{n-1} = \begin{pmatrix} 1.2 \\ 1 \end{pmatrix}, \mathbf{x}_n = \begin{pmatrix} 1.15 \\ 0.98 \end{pmatrix}.$$

Calculate  $\hat{\mathbf{x}}_{n+1}$ ,  $\hat{\mathbf{x}}_{n+2}$  and their respective error variance-covariances.

[5 + 16 = 21 marks]

5. In this question, a plot of the time series called GR is shown in Figure 8.

Figure 8: Plot of time series GR

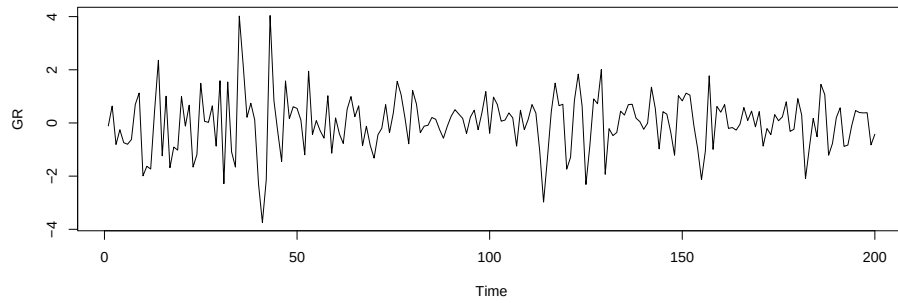


Figure 9: Autocorrelogram of time series GR

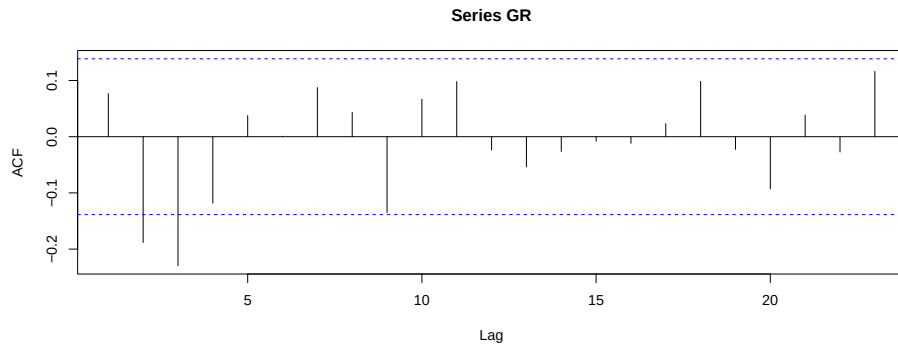
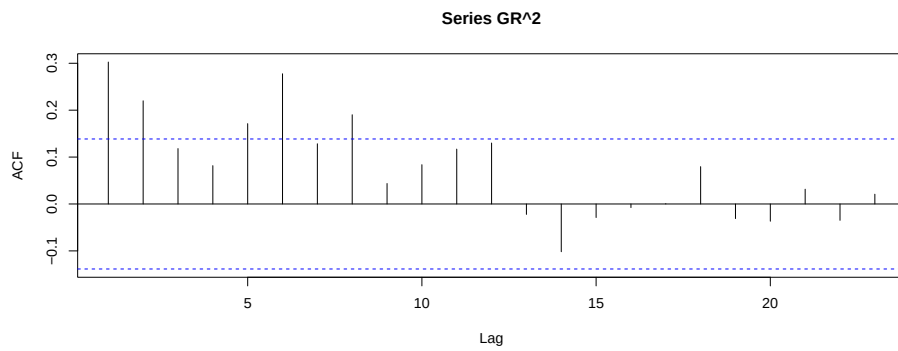


Figure 10: Autocorrelogram of square of time series GR



The autocorrelogram of the series and the square of the series are shown in Figures 9 and 10 respectively. The sample extended autocorrelogram chart of the square of the series is shown below.

```
AR/MA
  0 1 2 3 4 5 6 7 8 9 10 11 12 13
0  x x o o x x o x o o o  o  o  o
1  x o o o o x o x o o o  o  o  o
2  x x o o o x o o o o o  o  o  o
3  x o o o o x o o o o o  o  o  o
4  o x o o o x o o o o o  o  o  o
5  x x x x x o o o o o o  o  o  o
6  x x o o x x o o o o o  o  o  o
7  x x o o x o o x o o o  o  o  o
8  x o o o o o o x o o o  o  o  x
9  x o x x o o o x o o o  o  o  o
10 x o o o o o o x o x o  o  o  o
```

Table 3 lists the possible GARCH models that we are able to fit to the series without any convergence error and their respective AIC values. The square of the residuals of these model also do not show any significant spikes in their respective autocorrelograms.

Table 3: List of GARCH models and their AIC values

| GARCH | (0,1)    | (0,2)    | (0,3)    | (0,4)    |
|-------|----------|----------|----------|----------|
| AIC   | 556.3424 | 552.6545 | 550.9718 | 546.7815 |

- Does the time series GR look heteroscedastic or non-heteroscedastic? Provide your reasons.
- Looking at Figures 9 and 10, and the extended ACF chart of the square of GR, what are the possible initial models that you could try to fit to GR?
- The summary output for the GARCH(0,2) model is shown below.

Call:

```
garch(x = GR, order = c(0, 2))
```

Model:

```
GARCH(0,2)
```

Residuals:

```
      Min      1Q  Median      3Q      Max
-3.2037 -0.6069  0.1187  0.6548  2.7111
```

Coefficient(s):

|    | Estimate | Std. Error | t value | Pr(> t ) |     |
|----|----------|------------|---------|----------|-----|
| a0 | 0.40321  | 0.08738    | 4.614   | 3.94e-06 | *** |
| a1 | 0.56831  | 0.14316    | 3.970   | 7.20e-05 | *** |
| a2 | 0.19269  | 0.10936    | 1.762   | 0.0781   | .   |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Diagnostic Tests:

Jarque Bera Test

data: Residuals

X-squared = 3.8153, df = 2, p-value = 0.1484

Box-Ljung test

data: Squared.Residuals

X-squared = 0.049457, df = 1, p-value = 0.824

Write down the equation of this model.

[2 + 4 + 6 = 12 marks]

– END OF PAPER –